

CE 301: Strength of Materials and Structures

Course Code	CE 301
Course Name	Strength of Materials and Structures
L-T-P-C	3-0-2-4
Prerequisites	IC 240 - Mechanics of Rigid Bodies
Intended for	UG
Distribution	Discipline Core
Approval	9th Senate (May 2016)

Course Contents

- **Types of structures**
Free body diagram, conditions of equilibrium, statically determinate and indeterminate trusses, beams and frames. [2 Lectures]
- **Mechanics of small deformation**
Concepts of stress and strain, stress-strain characteristics of ductile and brittle materials, elastic constants and their relationships, thermal stresses. [12 Lectures]
- **Members subjected to flexural loads**
Shear force and bending moment in determinate beams. Calculation of deflection by double integration, moment-area and unit load methods. [12 Lectures]
- **Columns**
Euler's theory, Critical load for different end conditions, eccentric loading, columns with small initial curvature. [4 Lectures]
- **Influence lines for statically determinate structures**
Moving loads on beams and trusses; Maximum shear force and bending moment due to moving loads. [3 Lectures]
- **Analysis of indeterminate structures by slope-deflection method**
Statically indeterminate beams subjected to loads and uneven settlement of supports, Analysis of rigid frames with and without side sway. [9 Lectures]

Laboratory Modules

1. Study of reactions in beams with different support conditions.
2. Study of variation of bending moment and shearing force in a beam subjected to various loading conditions.
3. Study of load-deflection characteristics of determinate and indeterminate trusses.
4. Study of load-deflection characteristics of rectangular portal frames with uniform and non-uniform sections.
5. Study of stress and strains in the members of pin jointed frames.
6. Study of plastic bending of portal frames.
7. Load-deflection study of pinned and fixed arches.
8. Study of buckling characteristics of struts.
9. Study of horizontal and vertical deflections of asymmetric sections at various angles and loads.
10. Study of behavior of circular section under torsion.

Text Books

1. C. T. F. Ross, J. Case and L. Chivler, **Strength of Materials and Structures**, 4th edition, Butterworth Heinemann, 1999.
2. C.S.Reddy, **Basic Structural Analysis**, Tata McGraw Hill, 2001.
3. C.K. Wang, **Intermediate Structural Analysis**, Tata McGraw Hill, 2010.
4. R.C. Hibbeler, **Structural Analysis**, Pearson Education, 6th edition, New Delhi, 2008.
5. C.H. Norris, J.B. Wilbur, S.Utku, **Elementary Structural Analysis**, Tata McGraw Hill, 1991.
6. L. S. Negi and R. S. Jangjid, **Structural Analysis**, Tata McGraw Hill, 1997.

References

1. B. Onouye and K. Kane, **Statics and strength of materials for architecture and building construction**, 4th edition, Prentice Hall, USA, 2013.
2. G. Ranzi and I.B. Raymond, **Structural analysis: principles, methods and modelling**, CRC press, 2014.
3. A. Williams, **Structural analysis: in theory and practice**, Butterworth-Heinemann, UK and USA, 2009.
4. T.H.G. Megson, **Structural and stress analysis**, Butterworth-Heinemann, UK and USA, 2014.